THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A 5-digit number ends with digit 9 and has digit sum 35; the sum of its other four digits is: 1

(a) 35 **(b)** 44 **(c)** 17 **(d)** 26

2. The number of digits in the largest palindrome smaller than 100 is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

3. On reversing and adding 349, the first sum is: 1

(a) 1292 **(b)** 1281 **(c)** 1391 **(d)** 1303

4. 79 is added to its reverse and the sum is 176. How many reverse-and-add steps in all are needed to reach a palindrome? 1

(a) 6 **(b)** 8 **(c)** 5 **(d)** 7

5. How many steps does the Kaprekar routine take to reach 6174 from 9031? 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 2

6. The hailstone (Collatz) sequence of 22 has how many terms (including 22 and 1)? 1

(a) 16 **(b)** 17 **(c)** 15 **(d)** 14

7. The sum of 5 consecutive counting numbers starting from 100 is: 1

(a) 510 **(b)** 500 **(c)** 515 **(d)** 525

8. A 5-digit number ends with digit 6 and has digit sum 24; the sum of its other four digits is: 1

(a) 27 **(b)** 9 **(c)** 30 **(d)** 18

9. **Assertion (A):** The digit sum of 123 is 6. 1
Reason (R): 123 is divisible by 3.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** In a supercell pyramid with bottom row 3, 5, 2 the top number is 15. 1
Reason (R): Each number in a supercell pyramid is the sum of the two numbers just below it.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A number has 11 digits, all equal to 4. Find its digit sum, and say whether the number is divisible by 3 and whether it is divisible by 9. 2

12. How many steps does the Kaprekar routine need for 8532? Write the chain. 2

13. Find the number of terms in the hailstone (Collatz) sequence of 30 and its largest term. 2

14. A number has 15 digits, all equal to 6. Find its digit sum, and say whether the number is divisible by 3 and whether it is divisible by 9. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find a 4-digit palindrome of the form $abba$ with $a = 5$ and $b = 2$. Show that it is divisible by 11. 3

16. A row of 7 numbers starts 4, 7 and each next number is the sum of the previous two. Find the 7th number and the sum of all 7 numbers. 3

17. How many 4-digit numbers can be formed from 3, 4, 7, 9 using each digit once? What is the largest and the smallest? What is their difference? 3

OR

Find a 4-digit palindrome of the form $abba$ with $a = 7$ and $b = 4$. Show that it is divisible by 11. 3

18. A row of 9 numbers starts 1, 5 and each next number is the sum of the previous two. Find the 9th number and the sum of all 9 numbers. 3

OR

How many 4-digit numbers can be formed from 2, 6, 7, 8 using each digit once? What is the largest and the smallest? What is their difference? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Arjun plays the reverse-and-add game. He takes a number, adds its reverse, and repeats until he gets a palindrome (a number that reads the same both ways). He starts with 57. 5

(i) What is the reverse of 57?

(ii) What is $57 + 75$?

(iii) How many reverse-and-add steps does 57 take to reach a palindrome?

(iv) Do the same for 68 and give the palindrome and the number of steps.

20. A palindromic number of 5 digits is formed as $abcba$ where $a = 6$, $b = 3$, $c = 9$. Write the number, find its digit sum, check if it is divisible by 9, and find the smallest number to add to make it divisible by 9. 5

OR

Find the number of steps the reverse-and-add process takes from 192 to reach a palindrome, write each sum, and state whether the final palindrome has an even or odd number of digits. 5

END OF QUESTION PAPER — **Total: 40 Marks**